

TASK 01 | Create a **Product** table in the MySQL database

- 1) Logon to your MySQL account [See “Guide to Access the Module Software” on Blackboard] and access your database called **yourWestminsterID_0**.
- 2) Create a **Product** table in MySQL containing 8 columns as shown on the logical ERD (see logical ERD file on Blackboard) and according to the specifications below.

Column	Type	Length/values	Default	Default value	Null	Index	Auto Increment (A_I)
prodId	INT	4	None		Untick	PRIMARY	Tick
prodName	VARCHAR	30	None		Untick	--	Untick
prodPicNameSmall	VARCHAR	100	None		Untick	--	Untick
prodPicNameLarge	VARCHAR	100	None		Untick	--	Untick
prodDescripShort	VARCHAR	1000	None		Tick	--	Untick
prodDescripLong	VARCHAR	3000	None		Tick	--	Untick
prodPrice	DECIMAL	6,2	None		Untick	--	Untick
prodQuantity	INT	4	None		Untick	--	Untick

1ST METHOD: Write SQL Code (if you have prior experience in SQL and/or taken the 5COSC002W Database Systems module in semester 1)

- a) Open a new page on your code editor and save it in your C:\Drive as **homteq_db_tables.sql**.
- b) Type the SQL DDL code to create the **Product** table as specified above. Remember to create a primary key constraint.

```
CREATE TABLE Product
(
) ;
```

- c) Paste the code in the SQL window in PHPMyAdmin and run it.

2ND METHOD: Use the PHPMyAdmin GUI (if you have no prior experience in SQL and not taken the 5COSC002W Database Systems module in semester 1)

- a) Select your database **w1234567_0** on the left hand-side. Under **Create new table** type in the **name Product** and enter **number of columns 8** and click Go.
- b) Create your **Product** table fields by entering the information specified above.
- c) In order to ensure referential integrity, make sure that **Storage Engine** is **InnoDB**.
- d) Click **Save**.

TASK 02 | Populate the **Product** table in the MySQL database

1ST METHOD: Write SQL Code (if you have prior experience in SQL and/or taken the 5COSC002W Database Systems module in semester 1)

- a) Access the **homteq_db_tables.sql** on your code editor.
- b) Type in SQL DDL to create **4 new records** in the **Product** table.

```
INSERT INTO Product (      )  
VALUES (      );
```

- c) Remember to omit the product id from your SQL INSERT INTO code as it will be automatically incremented.
- d) Go to an online retailer such as John Lewis, PC World or Argos and grab a product name, a short product description and a long product description. Include them in your SQL INSERT INTO code.
- e) Create an **images** folder in the **homteq** project folder in your C:\Drive.
- f) From the online retailer, grab a small picture and a large picture of your product. Save the actual images in your images folder. Include the image names in your SQL INSERT INTO code with the right file extension such as .jpg or .png.
- g) Include a price and a product quantity available in stock.
- h) Paste the code in the SQL window in PHPMysqlAdmin and run it.
- i) Repeat this process 4 times so that you have 4 products in your **Product** table.

2ND METHOD: Use the PHPMysqlAdmin GUI (if you have no prior experience in SQL and not taken the 5COSC002W Database Systems module in semester 1)

- a) Select the **Product** table on the navigation bar on the left hand-side.
- b) Click on the *insert* tab.
- c) Leave the **prodId** value blank as it will be auto incremented
- d) Go to an online retailer such as John Lewis, PC World or Argos and grab a product name, a short product description and a long product description. Include them in the correct fields of your table insertion GUI.
- e) Create an **images** folder in the **homteq** project folder in your C:\Drive.
- f) From the online retailer, grab a small picture and a large picture of your product. Save the actual images in your images folder. Include the image names in the correct fields of your table insertion GUI with the right file extension such as .jpg or .png.
- g) Add a price and a product quantity available in stock names in the correct fields of your table insertion GUI.
- h) After every new product click *Go*.
- i) Repeat this process 4 times so that you have 4 products in your **Product** table.

NOTE
SQL (Structured Query Language) is a language used to Create, Retrieve, Update and Delete records in a database (CRUD). MySQL is a database Management System.

TASK 03| Create a **db.php** database connection file

- 1) Open a new tab in your code editor, type in the following:

db.php

```
<?php
$dbhost = 'phpmyadmin.ecs.westminster.ac.uk';
$dbuser = 'yourWestminsterID';
$dbpass = 'yourMySQLRandomPwd';
$dbname = 'yourWestminsterID_0';

//create a DB connection
$conn = mysqli_connect($dbhost, $dbuser, $dbpass, $dbname);
//if the DB connection fails, display an error message and exit
if (!$conn)
{
    die('Could not connect: ' . mysqli_error($conn));
}
//select the database
mysqli_select_db($conn, $dbname);
?>
```

- 2) Save this file as **db.php** in the **C:\homteq**. When you save it, make sure you select *Save as type: PHP Hypertext Preprocessor file*.
- 3) Go to WinSCP and drag **db.php** from **C:\homteq** (left window pane) and drop it into the **public_html\homteq** folder (right window pane).

TASK 04| Create the **INDEX** page to display available product names and images (as dynamically retrieved from the database)

- 1) Go to your **C:\homteq** and, in the same folder, copy and paste your **template.php** file.
- 2) Rename the copy to **index.php**.
- 3) Open **index.php** in your code editor and modify it by inserting the sections of code that are highlighted in yellow (code) and green (comments).



Make sure you study and understand the code that you have entered (in yellow) alongside the comments (in green) as you will have to re-use this code in the following tutorial and you will be assessed on your understanding of this code.

index.php

```
<?php
include ("db.php"); //include db.php file to connect to DB

$pagename="Make your home smart"; //create and populate variable called $pagename
echo "<link rel=stylesheet type=text/css href=mystylesheet.css>";
echo "<title>".$pagename."</title>";

echo "<body>";
include ("headfile.html");
echo "<h4>".$pagename."</h4>";

//create a $SQL variable and populate it with a SQL statement that retrieves product details
$SQL="select prodId, prodName, prodPicNameSmall from Product";
//run SQL query for connected DB or exit and display error message
$exeSQL=mysqli_query($conn, $SQL) or die (mysqli_error($conn));

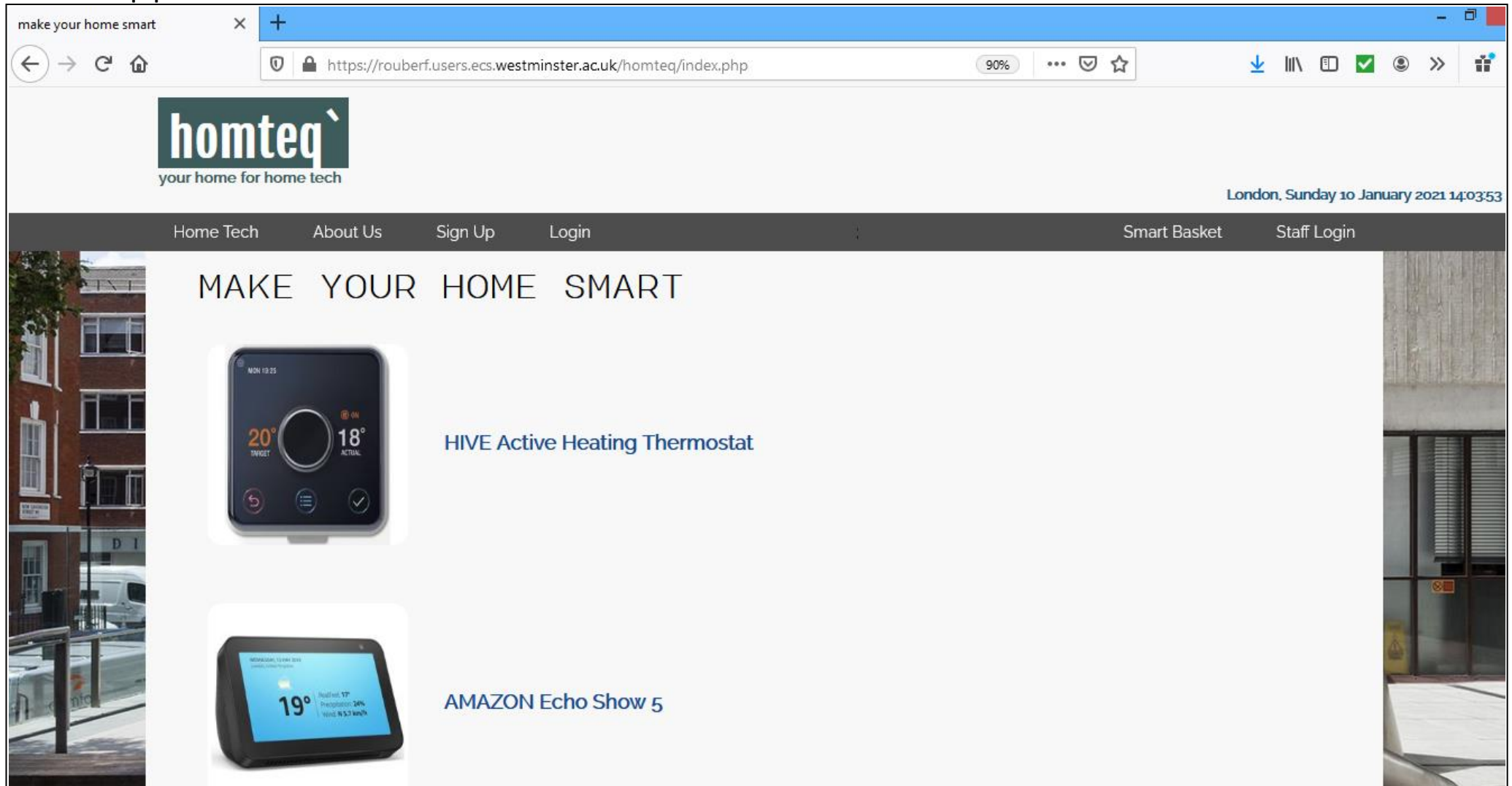
echo "<table style='border: 0px'>";
//create an array of records (2 dimensional variable) called $arrayp.
//populate it with the records retrieved by the SQL query previously executed.
//Iterate through the array i.e while the end of the array has not been reached, run through it
while ($arrayp=mysqli_fetch_array($exeSQL))
{
    echo "<tr>";
    echo "<td style='border: 0px'>";
    //display the small image whose name is contained in the array
    echo "<img src=images/'".$arrayp['prodPicNameSmall']."' height=200 width=200>";
    echo "</td>";
    echo "<td style='border: 0px'>";
    echo "<p><h5>".$arrayp['prodName']."'</h5>"; //display product name as contained in the array
    echo "</td>";
    echo "</tr>";
}
echo "</table>";
include ("footfile.html");
echo "</body>";
?>
```

- 4) Go to WinSCP and drag index.php from C:\homteq (left window pane) and drop it into the public_html\homteq folder (right window pane). Select 'Replace all'.

- 5) Go to WinSCP and also drag the **images** folder from **C:\homteq** (left window pane) and drop it into the **public_html\homteq** folder (right window pane). Select 'Replace all'.
- 6) Open a Web browser and type in **https://yourWestminsterID.users.ecs.westminster.ac.uk/homteq/index.php** with YOUR Westminster identifier.

You should be able to see a display similar to this:

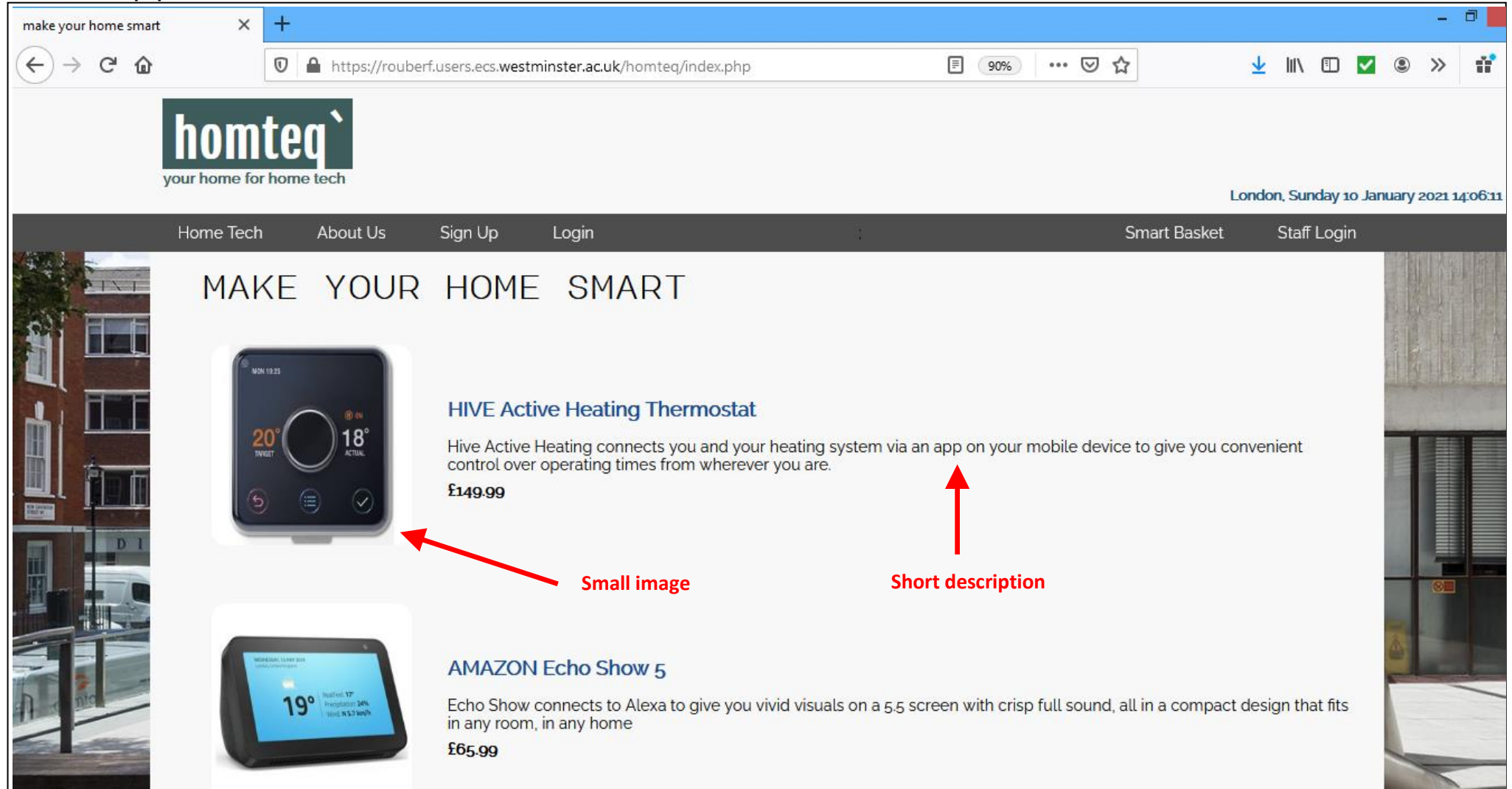
index.php



TASK 05 | Modify the **INDEX** page to also display available product descriptions and prices

- 1) Modify the code on **index.php** so that to also display, underneath the name of the product, a short product description and the product price (*hint: also edit SQL*)
- 2) Upload the modified **index.php** file using WinSCP and refresh your browser.
You should be able to see a display similar to this:

index.php



TASK 06 | Modify the **INDEX** page to make the images into clickable anchors that link to the **PRODUCT** page

- 1) Access the **index.php** file and locate the line of PHP code that displays the image using the **** tag.
- 2) Right before the image line, insert the following PHP code.
This creates an anchor tag **<a>** which specifies the HTML reference **prodbuy.php** to which the user is going to get directed when the user clicks on the anchor. It also specifies that the product id gets appended to the URL and passed to **prodbuy.php** using the GET method.
To do this, it uses a query string i.e a name/value pair which here is **prod_id** for the name and **\$arrayp['prodId']** for the value.

```
//make the image into an anchor to prodbuy.php and pass the product id by URL (the id from the array)
echo "<a href=prodbuy.php?u_prod_id=".$arrayp['prodId']. ">" ;
```

- 3) Right after the image line, close the anchor tag **** as follows.

```
//close the anchor
echo "</a>" ;
```

- 4) Upload the modified **index.php** file using WinSCP and refresh your browser.

You should be able to see a display identical to the previous one, but you should be able to click on each image and a link to the following URL should be activated, as illustrated on the display below.

https://yourWestminsterID.users.ecs.westminster.ac.uk/homteq/prodbuy.php (with your Westminster identifier).

make your home smart

https://rouberf.users.ecs.westminster.ac.uk/homteq/index.php

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MAKE YOUR HOME SMART



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https://rouberf.users.ecs.westminster.ac.uk/homteq/prodbuy.php?prod_id=2

Hover the image with mouse then click on it to activate the link to PRODUCT PAGE

When hovering the image with mouse, check that you can see the PHP page that is being linked to and the product ID